

Environmental Studies FACT SHEET



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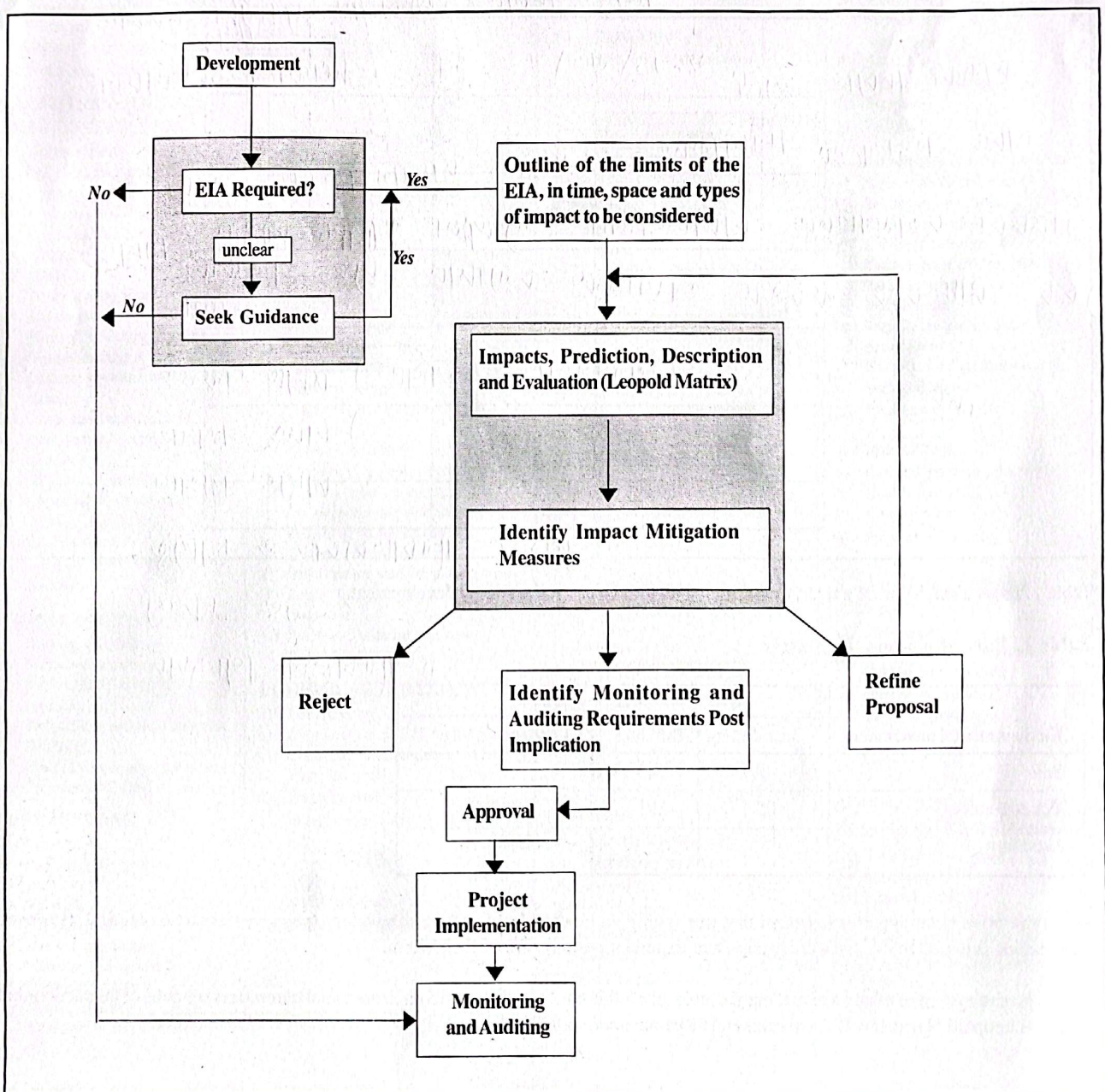
Number 101

The Leopold Matrix

This Factsheet explains how the Leopold Matrix is used in Environmental Impact Assessment to help us decide whether a major development should be allowed to go ahead.

Environmental Impact Assessments (EIAs) evaluate the harmful impacts that a potential development (new reservoir, new airport etc) might have on the environment. Fig 1. shows the main stages of an EIA.

Fig 1. EIA main stages



The Leopold Matrix

The Leopold Matrix was developed by the US Geological Service. It was designed for the assessment of impacts associated with almost any type of construction project. It considers 8,800 different possible interactions or impacts.

All of the possible activities associated with the project that might have a positive or negative impact e.g. building, forest clearance, gaseous emissions, noise are listed along the top of the matrix.

The existing environmental and social conditions that could possibly be affected by the project are put on the vertical axis divided into three major groups:

- physical conditions: soil, water, air...
- biological conditions: fauna, flora, ecosystems...
- social and cultural conditions: land use, historical and cultural issues, populations, economy

The cells where there is any possible interaction are divided into two by a slash.

In the top left hand corner of each cell a number from 1-10 is used to indicate the **magnitude** of the impact. A plus sign (+) is used to indicate **positive** impacts.

Similarly, In the bottom right hand corner of each cell a number 1-10 is used to indicate the **importance** of each possible impact (Fig 2).

Fig 2. Leopold Matrix

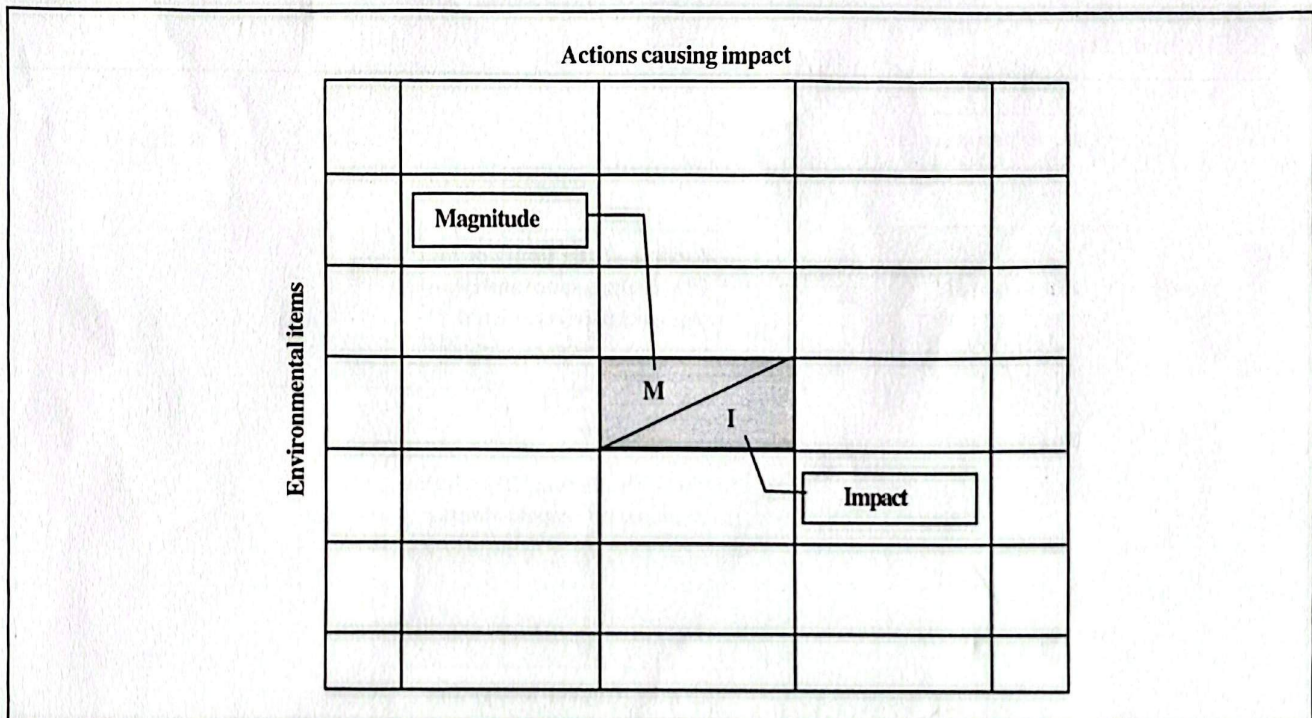


Table 1 shows a small part of a matrix used to predict the impact of a proposed development.

Table 1. Part of a Leopold matrix

Environmental parameter	Activities that cause impact		
	Tree-felling	Building	Landscaping
Soil	8/9	4/3	
Water quality	5/6	2/1	
Land form			+7/+7

So, in the example above, it is expected that tree felling will have a large and very important (negative) impact on the soil. However, landscaping is expected to have a fairly large and important positive effect of land form.

This is just a tiny part of what an actual matrix looks like – this has 3 activities and 3 environmental parameters = 9 cells or interactions but an actual Leopold Matrix has 100 activities and 88 parameters = 8,800 cells!

The actual activities and parameters used are listed below.

- Part I Lists the Project Actions (Arranged Horizontally in the Matrix);
- Part 2 Lists the Environmental 'Characteristics' and 'Conditions' (Arranged Vertically in the Matrix)

PART 1: Project Actions		PART 2: Environmental 'Characteristics' & 'Conditions'	
A. MODIFICATION OF REGIME a) Exotic flora or fauna introduction b) Biological Controls c) Modification of habitat d) Alteration of ground cover e) Alteration of ground-water hydrology f) Alteration of drainage g) River control and flow codification h) Canalization i) Irrigation j) Weather modification k) Burning l) Surface or paving m) Noise and vibration B. LAND TRANSFORMATION AND CONSTRUCTION a) Urbanization b) Industrial sites and buildings c) Airports d) Highways and bridges e) Roads and trails f) Railroads g) Cables and lifts h) Transmission lines, pipelines and corridors i) Barriers, including fencing j) Channel dredging & straightening k) Channel revetments l) Canals m) Dams and impoundments n) Piers, seawalls, marinas, & sea terminals o) Offshore structures p) Recreational structures q) Blasting and drilling r) Cut and fill s) Tunnels and underground structures C. RESOURCE EXTRACTION a) Blasting and drilling b) Surface excavation c) Sub-surface excavation and retorting d) Well drilling and fluid removal e) Dredging f) Clear cutting & other lumbering g) Commercial fishing & hunting D. PROCESSING a) Farming b) Ranching and grazing c) Feed lots d) Dairying e) Energy generation f) Mineral processing g) Metallurgical industry h) Chemical industry i) Textile industry j) Automobile and aircraft k) Oil refining l) Food m) Lumbering n) Pulp and paper o) Product storage	E. LAND ALTERATION a) Erosion control and terracing b) Mine sealing and waste control c) Strip mining rehabilitation d) Landscaping e) Harbour dredging f) Marsh fill and drainage F. RESOURCE RENEWAL a) Reforestation b) Wildlife stocking & management c) Ground-water recharge d) Fertilization application e) Waste recycling G. CHANGES IN TRAFFIC a) Railway b) Automobile c) Trucking d) Shipping e) Aircraft f) River and Canal traffic g) Pleasure boating h) Trails i) Cables and lifts j) Communication k) Pipeline H. WASTE EMPLACEMENT & TREATMENT a) Ocean dumping b) Landfill c) Emplacement of tailings, spoil and overburden d) Underground storage e) Junk disposal f) Oil-well flooding g) Deep-well emplacement h) Cooling-water discharge i) Municipal waste discharge including spray irrigation j) Liquid effluent discharge k) Stabilization and oxidation ponds l) Septic tanks, commercial & domestic m) Stack and exhaust emission n) Spent lubricants I. CHEMICAL TREATMENT a) Fertilization b) Chemical deicing of highways, etc c) Chemical stabilization of soil d) Weed control e) Insect control (pesticides) J. ACCIDENTS a) Explosions b) Spills and leaks c) Operational failure	A. PHYSICAL & CHEMICAL CHARACTERISTICS 1. Earth a) Mineral resources b) Construction material c) Soils d) Landform e) Force fields & background radiation f) Unique physical features 2. Water a) Surface b) Ocean c) Underground d) Quality e) Temperature g) Snow, Ice, & permafrost 3. Atmosphere a) Quality (gases, particulates) b) Climate (micro, macro) c) Temperature 4. Processes a) Floods b) Erosion c) Deposition (sedimentation, precipitation) d) Solution e) Sorption (ion exchange, complexing) f) Compaction and settling g) Stability (slides, slumps) h) Stress-strain (earthquake) f) Recharge i) Air movements B. BIOLOGICAL CONDITIONS 1. Flora a) Trees b) Shrubs c) Grass d) Crops e) Microflora f) Aquatic plants g) Endangered species h) Barriers i) Corridors 2. Fauna a) Birds b) Land animals including reptiles c) Fish & shellfish d) Benthic organisms e) Insects f) Microfauna g) Endangered species h) Barriers i) Corridors	C. CULTURAL FACTORS 1. Land use a) Wilderness & open spaces b) Wetlands c) Forestry d) Grazing e) Agriculture f) Residential g) Commercial h) Industrial i) Mining & quarrying 2. Recreation a) Hunting b) Fishing c) Boating d) Swimming e) Camping & hiking f) Picnicing g) Resorts 3. Aesthetics & Human Interest a) Scenic views and vistas b) Wilderness qualities c) Open space qualities d) Landscape design e) Unique physical features f) Parks & reserves g) Monuments h) Rare & unique species or ecosystems i) Historical or archaeological sites and objects j) Presence of misfits 4. Cultural Status a) Cultural patterns (life style) b) Health and safety c) Employment d) Population density 5. Man-Made Facilities and Activities a) Structures b) Transportation network (movement, access) c) Utility networks d) Waste disposal e) Barriers f) Corridors D. ECOLOGICAL RELATIONSHIPS SUCH AS: a) Salinization of water resources b) Eutrophication c) Disease-insect vectors d) Food chains e) Salinization of surficial material f) Brush encroachment g) Other